Case Study on Demonstration of Network Communication Models (OSI / TCP-IP) using a Network Simulation Tool

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Aim

To demonstrate how data travels through the layers of the OSI and TCP/IP models by building a small network in Cisco Packet Tracer and studying its packets in simulation mode.

Objectives

- Understand the seven OSI layers and the four TCP/IP layers and how they correspond.
- Build and configure a two-subnet network with a router, a switch, PCs and a server.
- Observe encapsulation and de-encapsulation of packets in simulation mode.
- Relate each observed protocol (ARP, ICMP, DNS, TCP, HTTP) to its layer.

Selected Network Scenario

Small office network (illustrative). An office has two departments on separate subnets. The Sales department uses PCs connected to a switch, and the company web server sits in a second subnet. A router connects the two. The task is to prove that a PC can reach the web server and to watch what happens, layer by layer, when it does.

Reference Models

The OSI model divides communication into seven layers so that each can be designed and troubleshot separately. The TCP/IP model used on the Internet groups the same functions into four layers. Packet Tracer displays events by OSI layer, which makes it a convenient teaching tool.

OSI layer	TCP/IP layer	Data unit	Example protocols	Device
7 Application	Application	Data	HTTP, DNS, DHCP	End device, server
6 Presentation	Application	Data	TLS, encoding formats	End device
5 Session	Application	Data	Session set-up and control	End device
4 Transport	Transport	Segment	TCP, UDP	End device
3 Network	Internet	Packet	IP, ICMP, ARP (boundary)	Router
2 Data Link	Network Access	Frame	Ethernet, MAC addressing	Switch
1 Physical	Network Access	Bits	Cables, signals	Hub, cable

Network Design

Device	Interface	IP address	Subnet mask	Default gateway
PC1 (Sales)	NIC	192.168.10.2	255.255.255.0	192.168.10.1
PC2 (Sales)	NIC	192.168.10.3	255.255.255.0	192.168.10.1
Router R1	Gig0/0	192.168.10.1	255.255.255.0	-
Router R1	Gig0/1	192.168.20.1	255.255.255.0	-
Web Server	NIC	192.168.20.10	255.255.255.0	192.168.20.1

Procedure

Step	Action
1	Open Packet Tracer and place two PCs, one switch, one router and one server; connect them with copper straight-through cables.
2	Assign the IP addresses from the design table in each device's configuration tab.
3	Configure the router interfaces using the commands shown below and bring them up.
4	Enable the HTTP service on the server and edit its home page text.
5	Test with ping from PC1 to the server, then open the server address in the PC's web browser.
6	Switch to **Simulation mode**, filter the events to ARP, ICMP, DNS, TCP and HTTP, and repeat the test while pressing the capture button.
7	Click each event and inspect the OSI Model tab, the inbound and outbound PDU details.

```
Router> enable
Router# configure terminal
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 192.168.20.1 255.255.255.0
Router(config-if)# no shutdown
```

Events Observed in Simulation Mode

Order	Event	Layer where it is decided	What it shows
1	ARP request and reply	2 / 3	PC finds the gateway's MAC address before sending off-network
2	ICMP echo request/reply	3	Reachability test; TTL decreases at the router
3	TCP SYN, SYN-ACK, ACK	4	Three-way handshake opens a reliable connection
4	HTTP GET and response	7	Browser requests the page; server returns it
5	TCP FIN / ACK	4	Connection is closed in an orderly way

Encapsulation Seen on the Outbound Path

Layer	Header added	Key fields seen
Application	HTTP request	Method GET, path, host
Transport	TCP header	Source port (random), destination port 80, sequence number
Network	IP header	Source 192.168.10.2, destination 192.168.20.10, TTL
Data link	Ethernet header	Source MAC of PC, destination MAC of router interface
Physical	Bits on the cable	Electrical signal

At the router the frame is received, the Ethernet header is removed (de-encapsulation up to layer 3), the destination network is looked up in the routing table, and a new Ethernet header is built for the next hop. The IP addresses stay the same from end to end, while the MAC addresses change at each hop. This is the most important observation of the experiment.

Switch versus Router

Aspect	Switch	Router
Decision based on	MAC address (layer 2)	IP address (layer 3)
Scope	Within one network	Between networks
Table used	MAC address table	Routing table

Troubleshooting Notes

- If ping fails, check the default gateway on the PC first, then the router interface status.
- A red link light means the cable type or interface is wrong, or the interface is shut down.
- If the web page does not open, confirm that the HTTP service is switched on in the server.

Results

- The PC reached the server across two subnets and the HTTP page was displayed.
- Each protocol event could be matched to its layer, confirming the theory of the OSI and TCP/IP models.
- MAC addresses changed hop by hop, while IP addresses remained constant.

Conclusion

Simulation tools make invisible network behaviour visible. By watching a single web request travel through the layers, the learner sees why layering works: each layer solves one problem and relies on the layer below. The OSI model is the better teaching reference, while the TCP/IP model describes what is actually used on the Internet.

Further reading: Cisco Networking Academy material for Packet Tracer; browser vendor documentation for developer tools (Chrome DevTools Network panel); IETF specifications for TCP, IP and HTTP. The office network, the web application and all timing or size values in this report are illustrative and assumed to demonstrate the method; they should be replaced with the values you observe in your own practical.